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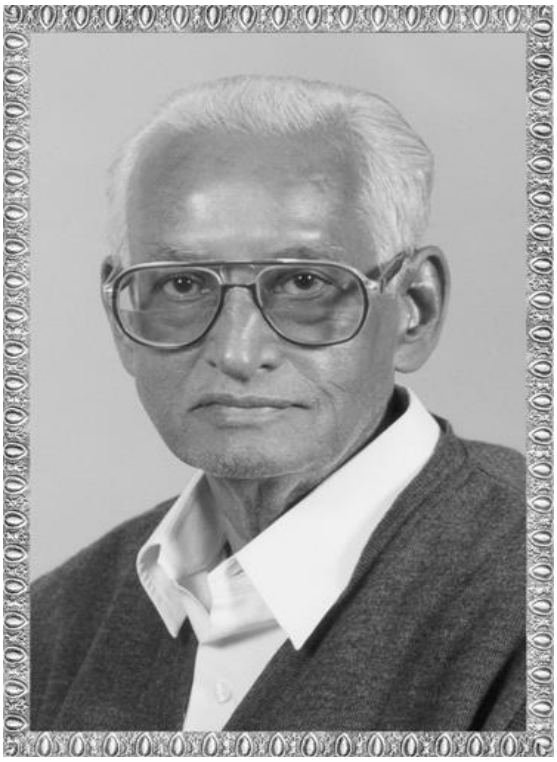

Lt. Shree Chimanbhai Shukla

BCA/BSCIT SEM-6 ASP.NET

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"SKY IS THE LIMIT"

SUBJECT: ASP.NET

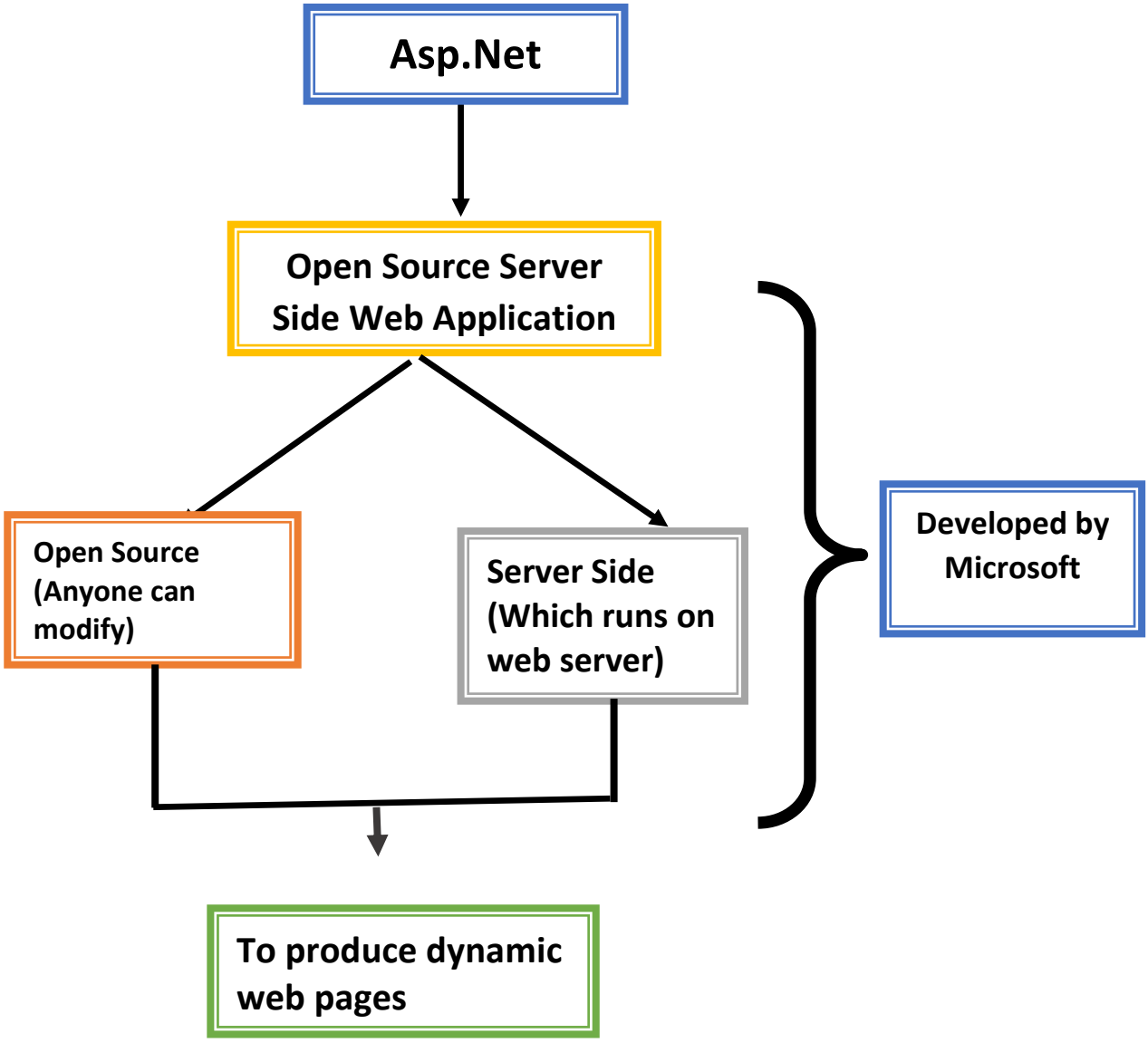
CHAPTER:1

Sr No.	Topic
1.	What is Asp.Net?
2.	Explain overview of asp.net framework
3.	Application Web Server
4.	Installation of IIS Server
5.	Types of files in asp.net
6.	Types of controls in asp.net
7.	Page Architecture
8.	Web form and features
9.	Introduction to standard controls (Button, Textbox, Checkbox, Label, Panel, Listbox, Dropdownlist etc.)
10.	File Upload Control
11.	What is Validation?
12.	Client Side Validation
13.	Server Side Validation
14.	Types (Required Field Validator, Range Validator, Compare Field Validator, Regular Expression Validator, Validation Summary Control)

Topic: What is Asp.Net?

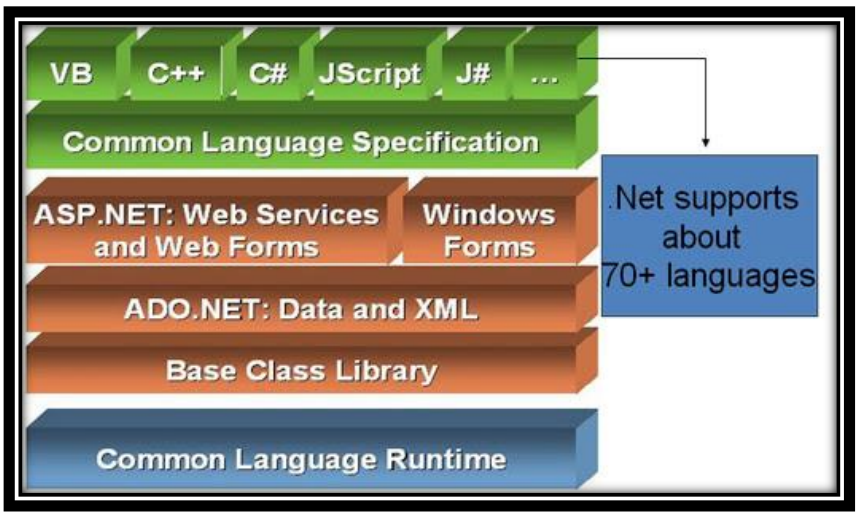
Ans:

- Asp.Net (Active Server Page) is open source server side web application.
- Asp is programming language and .Net is the framework.
- Framework means the platform in which different programming languages are executed.



Topic: Explain overview of Asp.Net Framework

Ans:



1) **.Net Languages:**
   - Net applications codes can be written in any of the languages.
   - Different 70 over languages are there.
2) **Common Language Specification (CLS)**
   - CLS defines set of rules and regulations that every programming languages running on .Net framework must follow
3) **Windows Form and Web Form:**
   - Two types of application asp.net supports:
     1) Windows Form: Contain the graphical representation of any window displayed in the application.
     2) Web Form: Contains only Coding Section
4) **Asp.Net Web Service:**
   - Asp.Net provides Services such as memory management, thread management, OOP features, Exception Handling.

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5) **ADO.NET: Data and XML:**
   - ADO.NET (Active X Data Object) is the technology used for working with data and database.
6) **BCL/FCL (Base Class Library/Framework Class Library)**
   - FCL is the collection of classes or namespaces that can be used to develop applications.
   - FCL is similar to Harder Files in c or c++
7) **CLR (Common Language Runtime)**
   - It is a runtime environment that manages and executes the code written in any .Net programming language.
   - It converts the code of different programming languages in to common language.

1 word Question Answer

Sr No.	Question	Answer
1.	CLR stands for	Common Language Runtime
2.	FCL Stands for?	Framework Class Library
3.	FCL is similar to header files in C or C++ (True/False)	True

Topic: Explain Characteristics of .Net Framework?

Ans:

Characteristics of .Net framework are:

1) **Multilanguage Development:**
   .Net Framework supports multiple languages like c#.net, asp.net etc. that is single framework can execute multiple languages.
2) **Platform and processor independent:**
   .Net framework is platform and processor independent which means that languages supported by .net framework can be executed on any type of Operating System and processor.
3) **Security:**
   .Net framework is secure because the code written is converted into CLR (Common Language Runtime)
4) **OOP:**
   OOP stands for Object Oriented Programming. .Net framework supports the concept of class and all other concepts of OOP.

1 word Question Answer

Sr No.	Question	Answer
1.	Asp.Net was developed by?	Microsoft
2.	OOP Stands for?	Object Oriented Programming
3.	ASP Stands for?	ActiveServerPage

Topic: Explain Client/Server Architecture

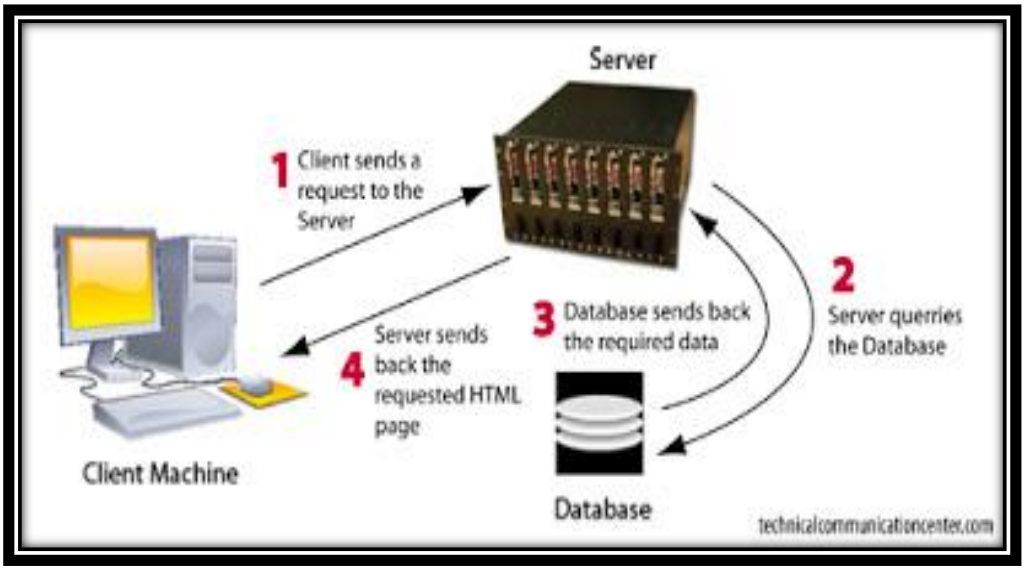
- Client Server architecture means the architecture having client and server.
- There are 2 types of client server architecture:
  - 2-tier architecture
  - 3-tier architecture

1) 2-tier architecture:

- 2-tier architecture is also known as traditional client server architecture
- In 2-tier architecture, client directly communicates with server

Working:

- In 2-tier architecture, clients sends the request to the server, server forward the client request to its database.



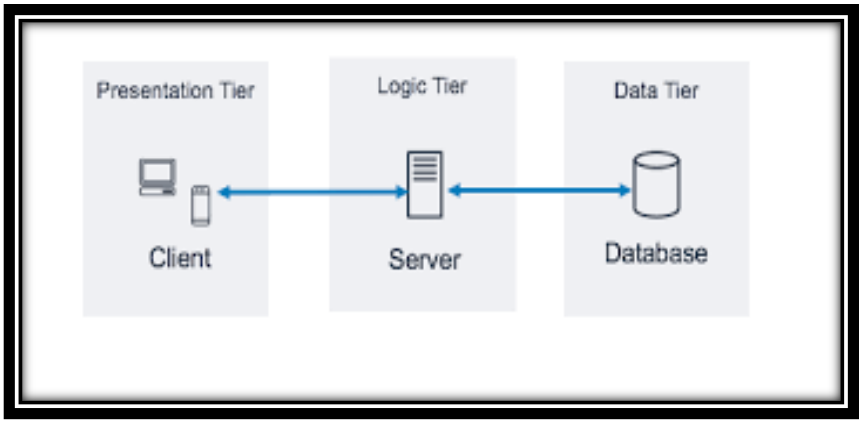
- Database checks if the requested data is available or not. If the data is Available, it response the server with the data and server response to the client with data.
- If the data requested by the client is not available in database, no response will be given.

- The **main disadvantage** of 2-tier architecture is that: Load on the server gets increased as client directly communicates with server and in the case if the server gets failed or crash, all the data of client will be lost.
- To overcome, this disadvantage, 3-tier architecture is used

3-tier architecture:

- In 3-tier architecture, the processing tier acts as mediator between application tier and data tier.
- Processing tier handles all the incoming requests of the client.
- Client cannot directly communicate with server

Working:



- In 3-tier architecture, client (presentation tier) sends the request to the processing tier/logic tier.
- Logic tier forwards the client request to the data tier (database) and searches for the data that client have requested.
- If the data is available, data tier response to the logic tier and logic tier forwards to the presentation tier.

- The **main advantage** of this architecture is that: load on the server does not get increased as processing tier handles all the interactions between client and server.
- In the case, if the server crash or fails, the data of client will not be lost as all the client requests are handled by processing or logic tier

1 word Question Answer

Sr No.	Question	Answer
1.	The architecture having only client and server is known as?	2-tier
2.	In ……architecture, processing tier handles the interaction between browser client and database tier	3-tier

Topic: Explain Application web server and how to execute multiform in web application OR Explain IIS

Ans:

Web Server:

- Web server delivers static content like HTML Page, images, Videos and files
- Web server respond only to the web based HTTP request.
- It can not access Database
- It is intended to be used for static content only.

Application Server:

- Application server deliver dynamic content like real time updates, personalized information and custom support.
- It not only respond to web based request but also delivering dynamic content.

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- It can access Database
- It is intended for dynamic content.

IIS:

- IIS stands for Internet Information Server/Service
- IIS is one of the most popular web servers from Microsoft that is used to host and provide internet based services to Asp.Net and Asp Web applications
- IIS is a Microsoft web server that runs on windows operating system and is used to exchange static and dynamic web content with internet users.
- IIS can be used to host, deploy and manage web applications using Asp.Net
- Although, Asp.Net development server comes inbuilt with visual studio, you can install IIS for better scalability of your web application after you upload it on real web server
- If you want to install IIS, you must have windows XP CD with you as IIS pack comes inbuilt as part of XP

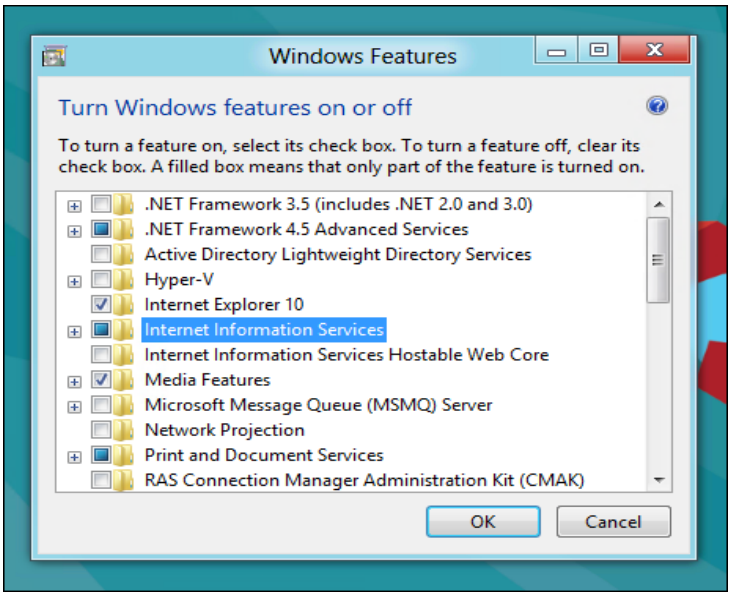
Steps for installation:

Step 1: Go to start

Step 2: Start -> Control Panel

Step 3: Control Panel-> Add/Remove Programs ->

Add/Remove Windows Components-> IIS



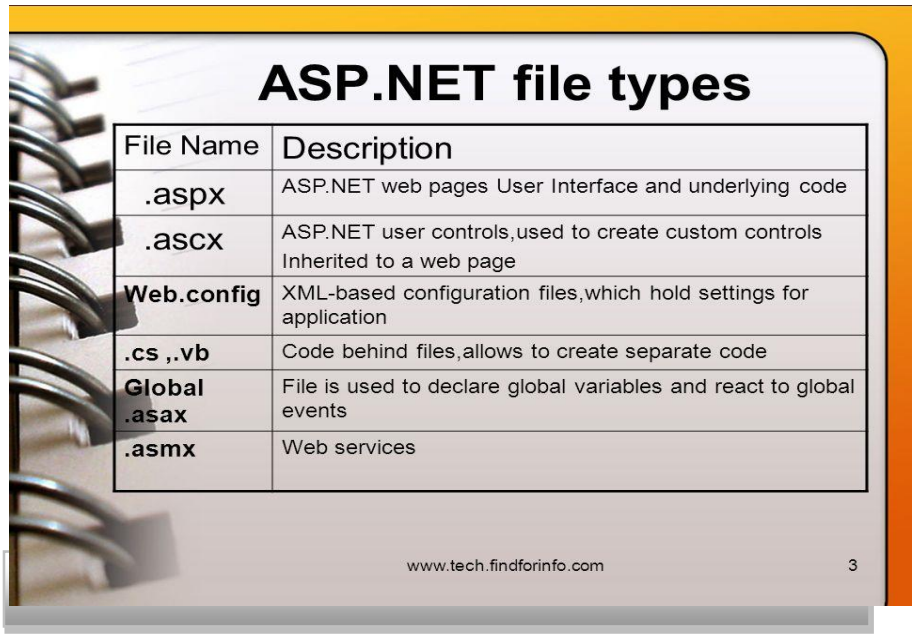
Execution of Multi Form in web application:

- Asp.Net relies on SFI (Single Form Interface) instead of MFI because SFI is a key element for implementation of Asp.Net state maintenance feature.
- So, if you host two or more server side form you will get runtime error.
- To execute multiple forms in web application, we need to use MVC (Model View Controller).

1 word Question Answer

Sr No.	Question	Answer
1.	IIS stands for?	InternetInformation Server
2.	In order to install IIS, you must have ….	Windows XP CD.

Topic: Explain types of files used in asp.net



ASP.NET file types

File Name	Description
.aspx	ASP.NET web pages User Interface and underlying code
.ascx	ASP.NET user controls,used to create custom controls Inherited to a web page
Web.config	XML-based configuration files,which hold settings for application
.cs ,.vb	Code behind files,allows to create separate code
Global .asax	File is used to declare global variables and react to global events
.asmx	Web services

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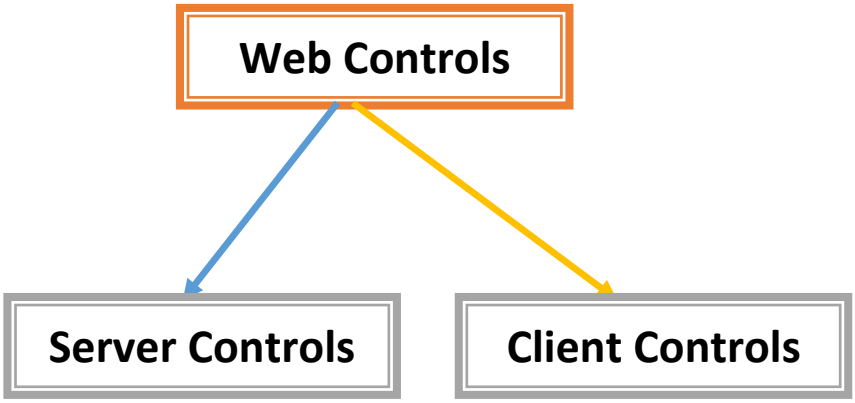
1) **.aspx:** It is the extension of any .Net file or Webfom. They contain user interface and underlying application code.
2) **.ascx:** These are Asp.Net user controls. User controls cannot be accessed directly but they should be used on web page
3) **Web.Config:** It is XML based configuration file which includes settings for security, state management, memory management etc.
4) **.cs/.vb:** These are code behind files that contain c# code pr vb code.
5) **global.asax:** It is the global application file and you can use this file to define global variables.
6) **.asmx:** These are asp.net web services files

1 word Question Answer

Sr No.	Question	Answer
1.	……are the code behind files that are used to create separate code	.cs/.vb
2.	……..are the asp.net web services file	.asmx
3.	…….is the XML based configuration file	Web.config

Topic: Explain types of controls in asp.net



1) **Server Controls:**
   - These are the controls which execute on server side. The controls which execute on server side have two important properties
   - Server controls provides automatic state management

runat="Server"
ID

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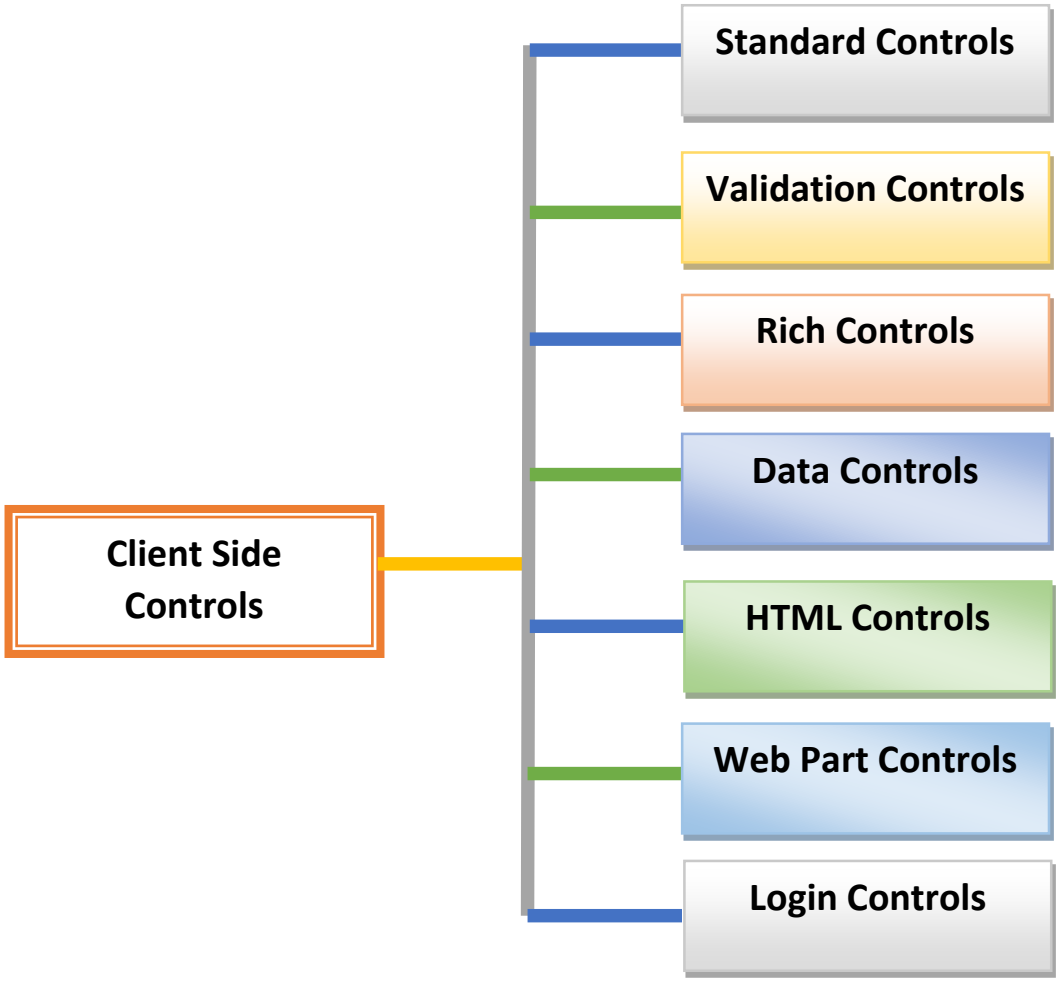
2) **Client Controls:**

- These are the controls which don't execute on server side. Generally, HTML Controls are known as client controls.
-



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1) **Standard Controls:**
   - This type of controls includes buttons, input fields, label, textbox, checkbox etc.
2) **Validation Controls**:
   - This type of controls enables you to validate form data before you submit data to server.

Example: RequiredFieldValidator

3) **Rich Controls:**
   - These are the advanced controls which have ability to generate large amount of HTML markup and javascript.

Example: Calendar, FileUpload, AdRotator

4) **Data Controls:**
   - These controls enable you to work with data such as database data.

Example: You can use this controls to submit new records to database table or display list of database records.

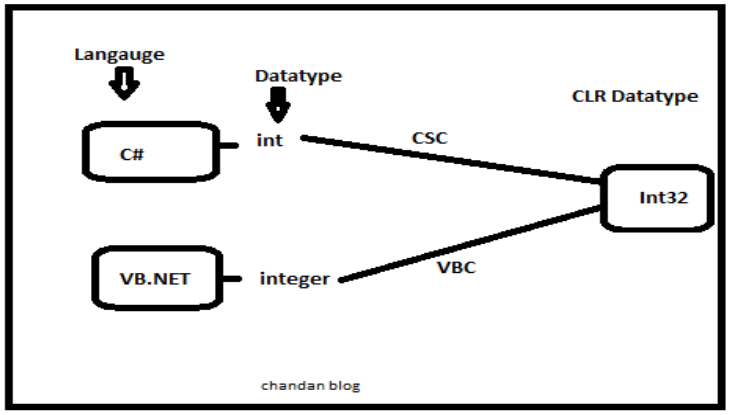
5) **HTML Controls:**
   - These controls enable you to convert any HTML tag into server side control.
6) **Web Part Controls:**
   - These controls enable you to build personalize portal applications
7) **Login Controls:**

These controls enable you to display login, change password and registration forms. Login control support form authentication.

1 word Question Answer

Sr No.	Question	Answer
1.	…… control is used to validate form	Validation
2.	Label and button are example of…	Standard Controls
3.	AdRotator is the example of ……	Rich Controls

Topic: What is CTS?

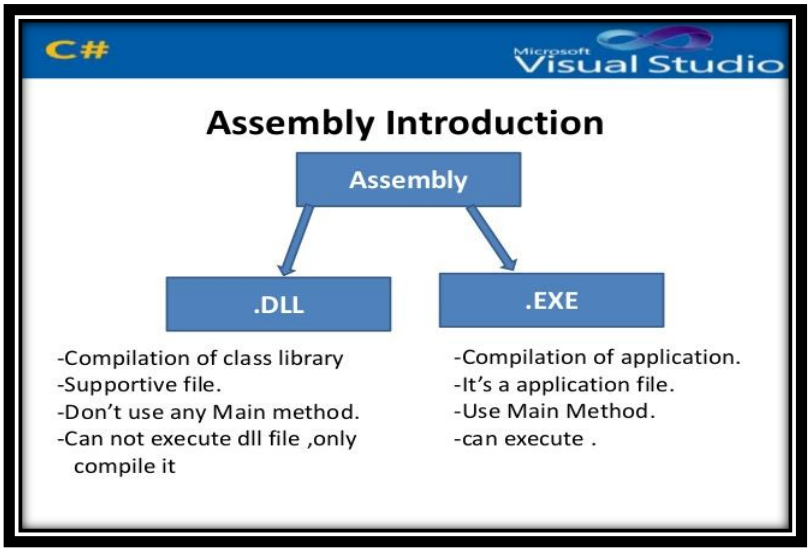


- CTS stand for Common Type System.
- CTS deal with data type. So we have several languages and each and every language has its own data type and one language data type cannot be understanble by other languages but .Net framework language can understand all the data types.
- In the above diagram, C# has int data type and vb.Net has integer data type. Hence a variable declared as an int in c# and integer in Vb.Net finally after compilation, uses same structure Int32 from CTS

1 word Question Answer

Sr No.	Question	Answer
1.	CTS stands for	Common Type System
2.	CTS provides Cross Language Integration? (True/False)	True

Topic: What is Assembly?



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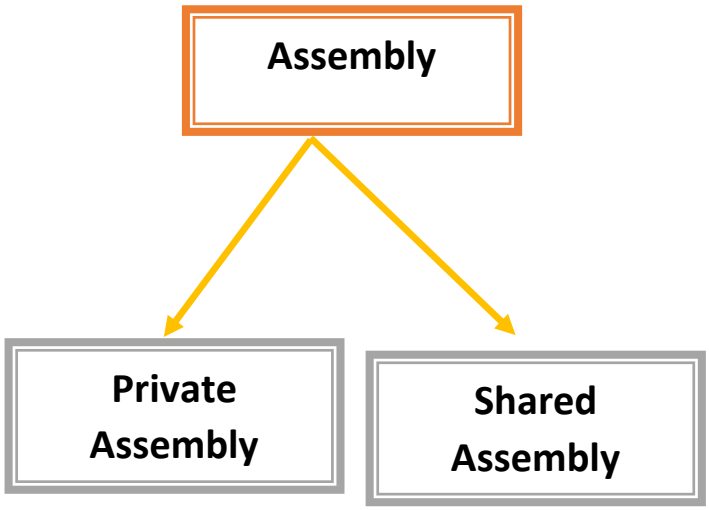
- An assembly is primary building block of .Net framework application.
- It is a single unit of deployment.



1) Private Assembly:

- It is used only by a single application and is stored in that applications installed directory.
- Private assembly requires us to copy separately in all application folders where we want to use that assemblies functionalities
- Without copying, we cannot access the private assembly Features.

2) Shared Assembly:

- This assembly can be referenced by more than one application.
- Shared assembly is not required to copy separately into all application folders. Shared assembly is also called public Assembly.
- Only one copy is required in system level, there is no need to copy the assembly into the application folder.
- Shared assemblies (also called strong named assemblies) are copied to a single location (usually the Global assembly cache)

1 word Question Answer

Sr No.	Question	Answer
1.	….is the primary building block of .Net framework	Assembly
2.	…..is used by single application	Private Assembly
3.	Which assembly is not required to copy separately in to all application folders	Shared Assembly
4.	Shared Assembly is also known as….	Public Assembly

Topic: Explain Metadata in Asp.Net

- Metadata in .Net is binary information which describes the characteristics of a resource.
- This information include Description of the Assembly , Data Types and members with their declarations and implementations, references to other types and members , Security permissions etc.
- ***During the compile time Metadata created with Microsoft Intermediate Language (MSIL) and stored in a file called a Manifest.***
- Both Metadata and Microsoft Intermediate Language (MSIL) together wrapped in a Portable Executable (PE) file.
- During the runtime of a program Just in Time (JIT) compiler of the Common Language Runtime (CLR) uses the Metadata and converts Microsoft Intermediate Language (MSIL) into native code.
- When code is executed, the runtime loads metadata into memory and references it to discover information about your code's classes, members, inheritance, and so on.

1 word Question Answer

Sr No.	Question	Answer
1.	…..in .Net is the binary information which describes characteristics of a resource	MetaData
2.	Meta Data created with MSIL is stored in file called?	Manifest
3.	Both Meta Data and MSIL are wrapped in single File known as …..	PE (Portable Executable)
4.	JIT Stands for?	Just In Time Compiler

Topic: Explain Namespace in Asp.Net

- Namespace is the Logical group of types or we can say namespace is a container (e.g Class, Structures, Interfaces, Enumerations, Delegates etc.),
- Example System.IO logically groups input output related features, System.Data.SqlClient is the logical group of ado.net Connectivity with Sql server related features.
- Namespaces allow you to create a system to organize your code.

Example:-

```
// Namespace Declaration
using System;
```

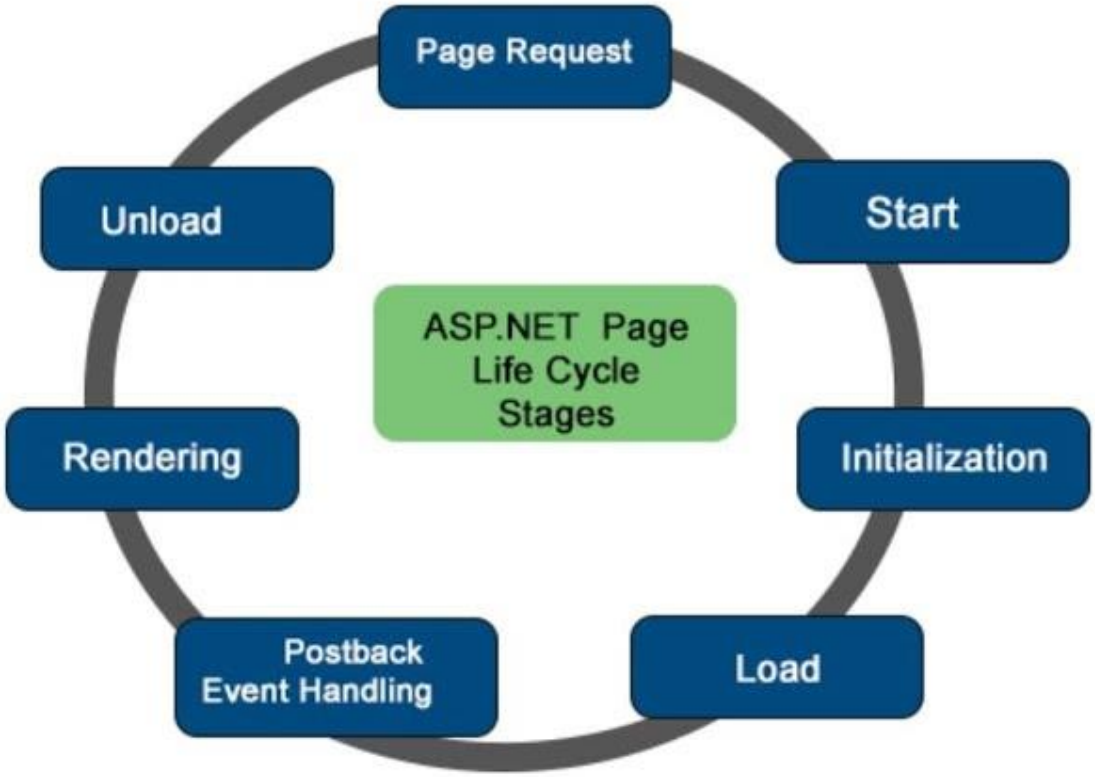
1 word Question Answer

Sr No.	Question	Answer
1.	……..is a logical container	Namespace
2.	……allows you to create a system to organize your code	Namespace

Topic: Explain Asp.Net Page Architecture

Ans:

- The architecture of an Asp.Net page is fundamentally based on layered structure and operates within the .Net framework
- When the user requests an ASP.Net page, the application processes it through a specific sequence of events known as page life cycle. This ensures that the page and its components are initialized, processed and rendered correctly.



- The main stages of Asp.Net page life cycle are:
  1) **Page Request:** Asp.Net decides whether to process the page from scratch or serve a cached version
  2) **Initialization:** Request and Response object are set and properties like IsPostBack are established. Dynamic controls and master pages are applied.
  3) **Load:** Control properties are loaded using data from the View state and control state. This is a key stage for restoring the state of the page across requests
  4) **Postback Event Handling:** If the request is a postback, eg. button click, the relevant event handler code is executed.
  5) **Rendering:** The Render method is called for each control. The final HTML mark up is generated and written to the Response object and ready to be sent to the client's browser.
  6) **Unload:** The page is sent to the client, and all resources and references like database connections are freed as clean up is performed.

Topic: What is web form? Explain features of web form

Ans:

- Web form are the pages that your user request using their browser.
- Asp.Net web pages are the web forms that are executed on server using Asp.Net technology
- Asp.Net web form have 2 parts: 1) .aspx file and 2) Coding section

Features:

1) **Server Controls:** It provides a large set of server controls that are executed on server side. It can be same as HTML controls like textbox, checkbox etc

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2) **Master Page:** It is responsible for same or consistent layout of web applications. It gives proper appearance and standard to different pages.

3) **Routing:** URL is a user request that enters in browser to browse in a specific place. URL Routing is a technique that manages how URL are handled and accessed on website

4) **State Management:** This feature of web form includes many different state management techniques like session, cookie that preserve both a page and application basis

5) **Security:** Developing a secure application is most important aspect of software development process. Asp.Net web forms provides different validation controls.

6) **Styling:** Web forms are styled using CSS to match the website design and improve visual appeal. Example: Colors, Fonts, Layout

Topic: What is Auto Postback property?

Ans:

- The `AutoPostBack` property in ASP.NET Web Forms is a boolean property that, when set to `true`, causes a control to **automatically send the entire page back to the server** for processing immediately after a relevant client-side event occurs (e.g., changing a dropdown list selection or losing focus from a text box).

Topic: Explain Standard Controls

1) Label Control

- Whenever you need to modify the text displayed in a page dynamically, you can use the Label control.

Properties:

Sr No.	Property	Description
1.	BackColor	Enables you to change the background color of the label.
2.	BorderColor	Enables you to set the color of a border rendered around the label
3.	BorderStyle	Enables you to display a border around the label.
4.	Font	Enablesyou to set the label's font properties
5.	ForeColor	Enables you to set the color of the content rendered by the label

1 word Question Answer

Sr No.	Question	Answer
1.	**…….control is used to display or modify text**	**Label**
2.	**Which property is used to set the label's font?**	**Font**

2) TextBox Control

- The TextBox control can be used to display three different types of input fields depending on the value of its TextMode property.
- The TextMode property accepts the following three values:

1) SingleLine : Displays a single-line input field.
2) MultiLine : Displays a multi-line input field.
3) Password : Displays a single-line input field in which the text is hidden.

Properties:

Sr No.	Property	Description
1.	TabIndex	Enables you to specify the tab order of the textbox.
2.	Wrap	Enables you to specify whether text word wraps when the TextMode is set to Multiline.
3.	BorderStyle	Enables you to display a border around the label.
4.	Font	Enablesyou to set the label's font properties
5.	ForeColor	Enables you to set the color of the content rendered by the label

Events:

1) **Text Changed:** This event is raised on the server when the contents of the text box are changed.
2) **Focus:** This event enables you to set the initial form focus to the textbox.

1 word Question Answer

Sr No.	Question	Answer
1.	**……control is used to enter the text from the user**	**TextBox**
2.	**…….value of textmode property allows you to enter multiple lines in textbox**	**MultiLine**
3.	**……property is used to specify whether text word wrap when textmode property is set to multiline**	**Wrap**

3) CheckBox Control

- This control is used to accept the choice from user. It is used to display multiple choices from which user can select none of them or many or all of them.

Properties:

Sr No.	Property	Description
1.	Checked	Enables you to get or set whether the checkbox is checked.
2.	Enabled	Enables you to display the checkbox
3.	TabIndex	Enables you to specify the tab order of the checkbox
4.	Text	Enables you to provide a label for the checkbox
5.	TextAlign	Enables you to align the label for the checkbox.

Events:

1) **CheckedChanged:** This event is raised on the server when the checkbox is checked or unchecked.
2) **Focus:** This event enables you to set the initial form focus to the checkbox

1 word Question Answer

Sr No.	Question	Answer
1.	**…….control is used to display multiple choices from which user can select none of them or many or all of them**	**CheckBox**

2.	**Which event is fired when the checkbox is checked or unchecked**	**CheckedChanged**

4) RadioButton Control:

- This control is used in the group. Only 1 radiobutton in a group of RadioButton controls can be checked at a time.
- Radiobutton are mutually exclusive.

Properties:

Sr No.	Property	Description
1.	Checked	Enables you to get or set whether the radiobutton is checked.
2.	Enabled	Enables you to display the radiobutton
3.	TabIndex	Enables you to specify the tab order of the radiobutton control.
4.	Text	Enables you to provide a label for the radiobutton control
5.	TextAlign	Enables you to align the label for the radiobutton.
6.	GroupName	Enables you to group radiobutton controls.

Methods:

1) **DataBind:** This method binds a data source to the server control and all its child controls

Events:

1) **CheckedChanged:** This event is raised on the server when the radiobuton is checked or unchecked.

1 word Question Answer

Sr No.	Question	Answer
1.	……..control is also known as mutually exclusive	Radiobutton
2.	Which property is used to group radiobutton controls?	GroupName

5) Button Control:

- This control used to submit the data to the server. It works like a push button, when you click data is submitted to the server.

Properties:

Sr No.	Property	Description
1.	Enabled	Enables you to disable the button control
2.	PostBackUrl	Enables you to post a form to particular page
3.	TabIndex	Enables you to specify tab order of the button control
4.	Text	Enables you to label the button control

Methods:

1) **Focus:** This event enables you to set the initial form focus to the button control

Events:

1) **Click**: This event is raised when the button control is click.

1 word Question Answer

Sr No.	Question	Answer
1.	……..control is used to submit data to the server	Button
2.	Which event is fired when button is clicked?	Click

6) Dropdown list Control:

- It is web server control which is used to create HTML select component. It allows to select option from drop down list

Properties:

Sr No.	Property	Description
1.	Items	Allows you to add data in the drop down list
2.	SelectedItems.Text	Gives the text of the item that is selected in drop down list.
3.	SelectedItems.Value	Gives the value of the item that is selected in drop down list
4.	SelectedItems.Count	Counts total number of items in dropdown list
5.	Text	Name of the drop down list
6.	Font	Font of the dropdownlist text

Events:

1) SelectedIndexChanged: This event is fired or raised when dropdownlist item is selected or the index is changed

6) Listbox Control:

- It is a control that is used to store multiple items and allows user to select multiple items from listbox

Properties:

Sr No.	Property	Description
1.	Items	Allows you to add data in the listbox
2.	SelectedItems.Text	Gives the text of the item that is selected in lsitbox
3.	Selecttion Mode	Determines whether a user can select more than one list item
4.	SelectedItems.Count	Counts total number of items in listbox list

Events:

1) SelectedIndexChanged: This event is fired or raised when listbox item is selected or the index is changed

6) Link Button Control:

- The LinkButton control, like the Button control, enables you to post a form to the server.
- Unlike a Button control, however, the LinkButton control renders a link instead of a push button.

Properties:

Sr No.	Property	Description

1.	Enabled	Enables you to disable the Linkbutton control
2.	PostBackUrl	Enables you to post a form to particular page
3.	TabIndex	Enables you to specify tab order of the Linkbutton control
4.	Text	Enables you to label the Linkbutton control

Methods:

1) **Focus:** This event enables you to set the initial form focus to the Linkbutton control

Events:

1) **Click**: This event is raised when the Linkbutton control is clicked

1 word Question Answer

Sr No.	Question	Answer
1.	**….control renders a link instead of push button**	**LinkButton**
2.	**Which event is fired when linkbutton is clicked?**	**Click**

7) Image Button Control:

- The ImageButton control, like the Button and LinkButton controls, enables you to post a form to the server.
- However, the ImageButton control always displays an image

Sr No.	Property	Description
1.	ImageUrl	Enables you to specify URL to the image

2.	Alternate Text	Enables you to provide alternate text for the image
3.	TabIndex	Enables you to specify the tab order of the Image Button control.
4.	Description URL	Enables you to provide a link to the page that contains a detailed description of the image
5.	ImageAlign	Enables you to align the image relative to other HTML elements in the page.
6.	Enabled	Enables you to disable the ImageButton control

Methods:

1) **Focus:** This event enables you to set the initial form focus to the Image button control

Events:

1) **Click**: This event is raised when the Imagebutton control is clicked

1 word Question Answer

Sr No.	Question	Answer
1.	**……..property of image control specifies the URL of the image to be displayed**	**ImageURL**
2.	Which property enables you to provide a link to the page that contains a detailed description of the image.	**DescriptionURL**

8) Panel Control:

- The Panel control enables you to work with a group of ASP.NET controls. You can use a Panel control to hide or show a group of ASP.NET controls.

Sr No.	Property	Description
1.	Default Button	Enables you to specify the default button in a panel. The default button is invoked when you press the Enter button
2.	Grouping Text	Enables you to render the text of panel
3.	Scrollbars	Enables you to display scrollbars around the panel's contents.

1 word Question Answer

Sr No.	Question	Answer
1.	**……control enables you to work with a group of controls**	**Panel**
2.	**……property enables you to display scrollbars around the panel's content**	**Scrollbars**

9) Hyperlink Control:

- The HyperLink control enables you to create a link to a page. Unlike the LinkButton control, the HyperLink control does not submit a form to a server.

Properties:

Sr No.	Property	Description
1.	Enabled	Enables you to disable the hyperlink
2.	ImageURL	Enables you to specify the URL represented by hyperlink
3.	Text	Enables you to label the hyperlink

Note:

- You can specify an image for the HyperLink control by setting the ImageUrl property. If you set both the Text and ImageUrl properties, then the ImageUrl property takes precedence.

1 word Question Answer

Sr No.	Question	Answer
1.	**……control enables you to create a link to a page**	**Hyperlink Control**
2.	**……property enables you to specify the URL represented by hyperlink**	**ImageURL**

11) Image Map Control

- Map is a picture which is subdivided into areas. Image Map is also a single image but different parts of image are linked to different URLs.
- Example: you have Gujarat Map. When you click on different areas of Gujarat, the information of clicked city should be displayed.
- A Hotspot defines clickable region in image map.
- There are 3 different classes of Hotspot:

  - **Circle Hotspot**: Enables you to define circular region in image map.
  - **Polygon Hotspot**: Enables you to define irregularly shaped region in image map.

- **Rectangle Hotspot**: Enables you to define a rectangular region in image map.

- **Properties:**

Sr No.	Property	Description
1.	HotSpot	Enables you to retrieve the collection of Hotspot contained in Image Map Control
2.	Image URL	Enables you to specify the URL to the image
3.	Alternate Text	Enables you to provide alternate text for the image map.

1 word Question Answer

Sr No.	Question	Answer
1.	**……defines a clickable region in the Map.**	**Hotspot**
2.	**There are …….different classes of Hotspot.**	**3**

Topic: Explain File Upload Control

- It is an input controller which is used to upload file to the server. It creates a browse button on the form that pop up a window to select the file from the local machine.
- This is a server side control and ASP.NET provides own tag to create it.

Properties:

Sr No.	Property	Description
1.	BackColor	It is used to set the background color of the file upload control.

2.	Border Color	It is used to set the border color of the file upload control.
3.	Allow Multiple	It is used to allow upload multiple files by setting true or false.
4.	Font	It is used to set the font of the control.
5.	HasFile	Checks whether the file is selected or not

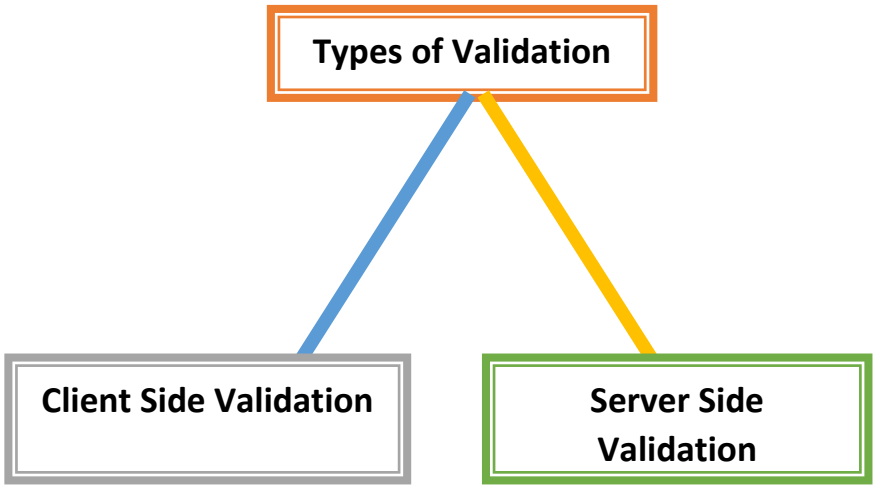
1 word Question Answer

Sr No.	Question	Answer
1.	**Which control is used to upload file?**	**File Upload Control**
2.	**Can File upload control upload multiple files?**	**Yes**

Topic: Explain types of validation.

Ans:

- Validation is a feature that checks user input data for accuracy and validity before it's processed by the web application



1) Client Side validation:

- It means checking of data is done at client end that is by the browser itself.
- The main theme behind developing the client side validation is fast response to the client.
- Various scripting languages can be used like javascript, DHTML for validation.
- Asp.Net provides several controls which are used for client side validation.

2) Server Side validation:

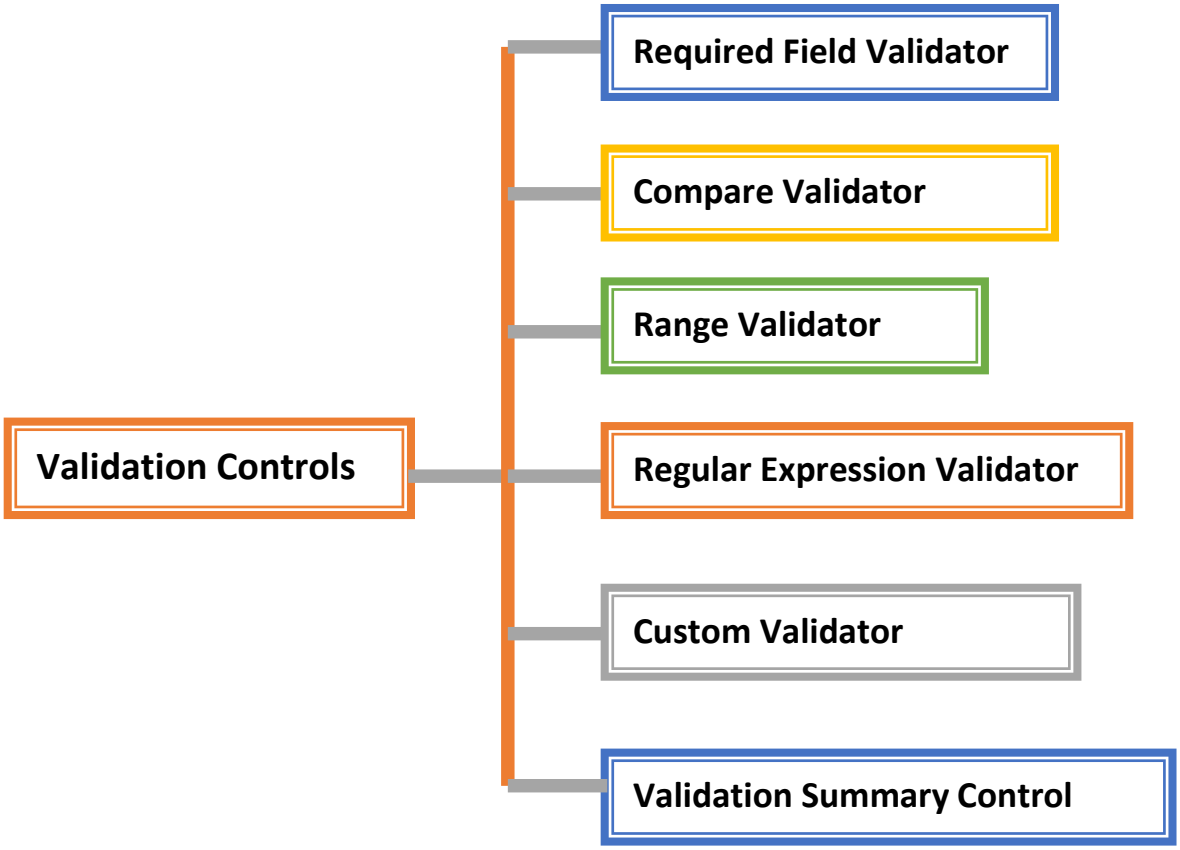
- It means checking of data is done at server side. Even though the data is checked at client side, it must be checked at server side. The reason behind this is attacker.
- Attacker might attack on your data and manipulate it or misuse it.
- Client side validation is done at browser level, but when the data is submitted to the server attacker/hacker might manipulate it.
- Server side validation is done using database programming tool. Asp.Net uses ADO.NET and related controls for this purpose.

1 word Question Answer

Sr No.	Question	Answer
1.	**Client side validation is done using …..**	**Javascript**
2.	**The reason to check or validate the data at server side is …..**	**Attacker/Hacker**

Topic: Explain types of validation controls.

Ans:



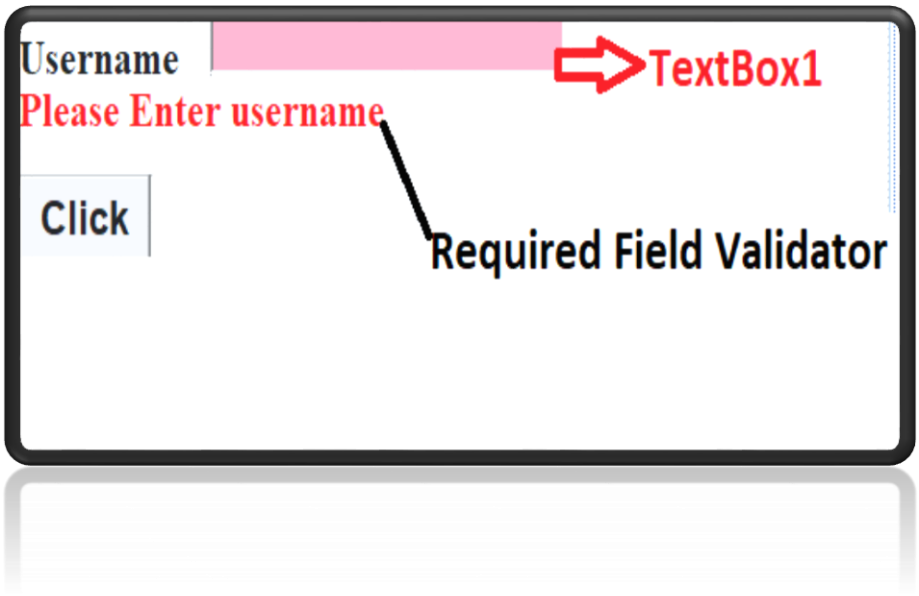
1) Required Field Validator:

- It is used to check whether the control has got some value or not. It forces user to enter the value for the control.

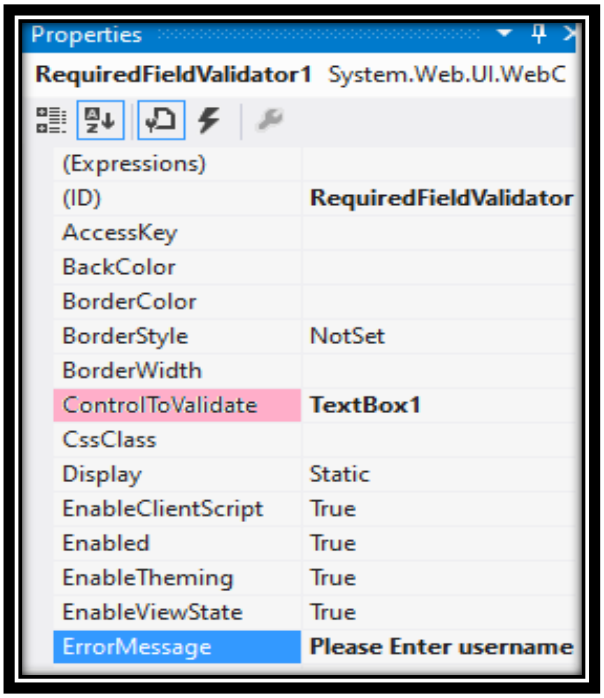
Sr No.	Property	Description
1.	Error Message	This property is used to set the message which is displayed when Validator control fails.
2.	Control to validate	This property is used to set the control which you want as required field.
3.	Text	to change the text of validation control

Example:

Design View:



Property Window:



1 word Question Answer

Sr No.	Question	Answer
1.	…….validation control is used when we want the field to be entered compulsory	Required Field
2.	Required field has ……property which shows the control on which validation is to be performed	Control to Validate

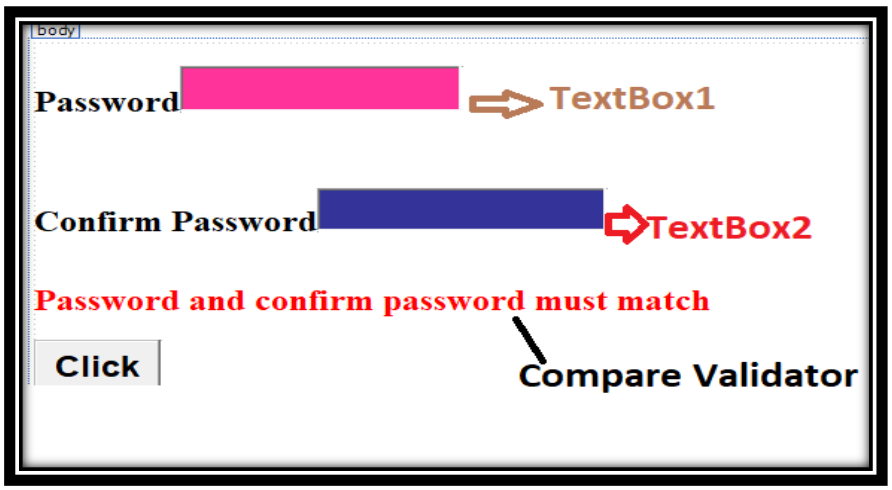
2) Compare Validator:

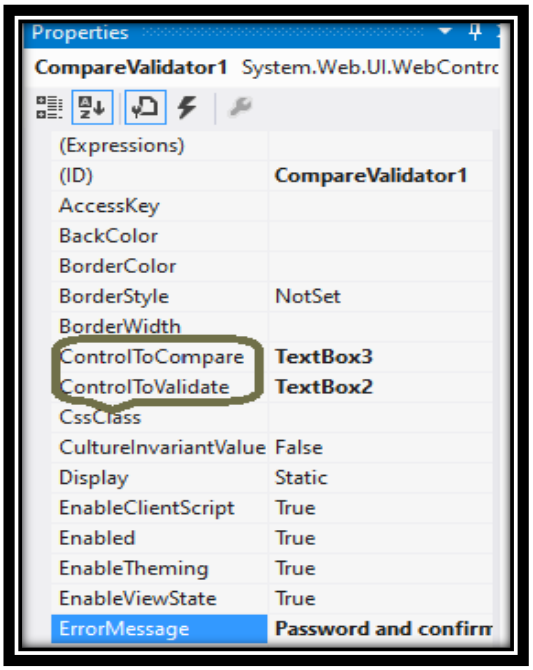
- This control is used to check value of two controls. It checks whether both controls have got same value or not. If both controls have got same value, it returns true else returns false.

Sr No.	Property	Description
1.	**Error Message**	**This property is used to set the message which is displayed when Validator control fails.**
2.	**Control to validate**	**This property is used to set the control which you want as required field.**
3.	**Control to Compare**	**This property is used to set the control which you want to compare.**
4.	**Text**	**to change the text of validation control**

Example:

Design View:





1 word Question Answer

Sr No.	Question	Answer
1.	……. validation is used to compare two different controls	Compare Validator
2.	……..is the example of compare Validator	Password and confirm password
3.	……..property indicates the control which is to be compared	Control to compare

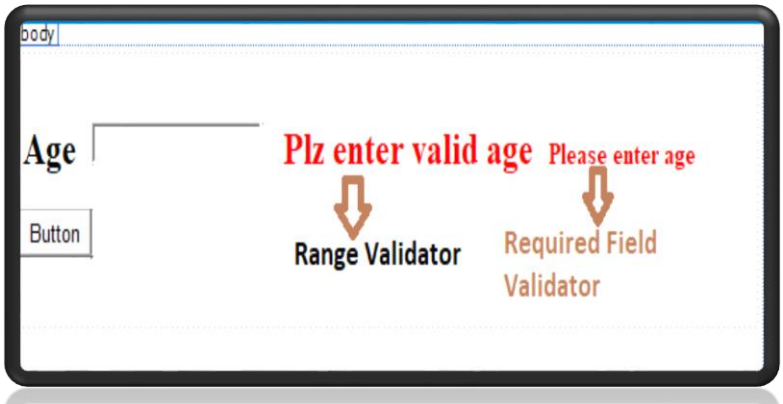
3) Range Validator:

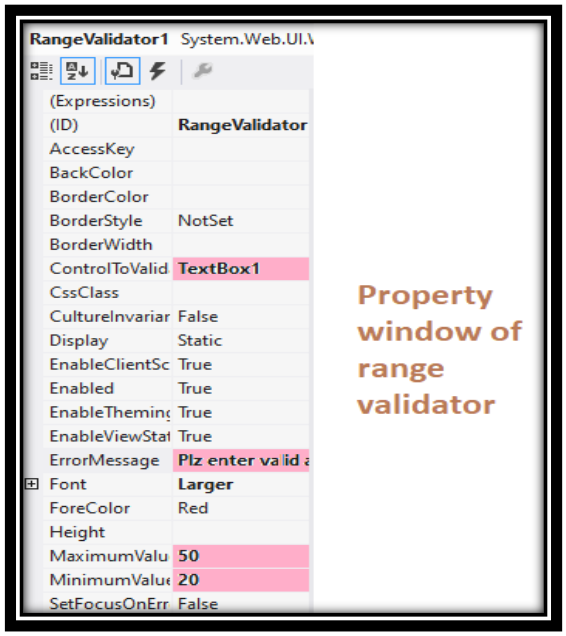
- This control is used to check that value of control is between specific range or not. The value which you enter into control can be numeric, date or string.
- Examples of range Validator are:
  - Input age within specific range

Sr No.	Property	Description
1.	Error Message	This property is used to set the message which is displayed when Validator control fails.
2.	Control to validate	This property is used to set the control which you want as required field.
3.	Minimum Value	This property is used to set the minimum value of the control to be validated
4.	Maximum Value	This property is used to set the maximum value of the control to be validated.
5.	Type	Specifies the type of the input field

Example

Design View:

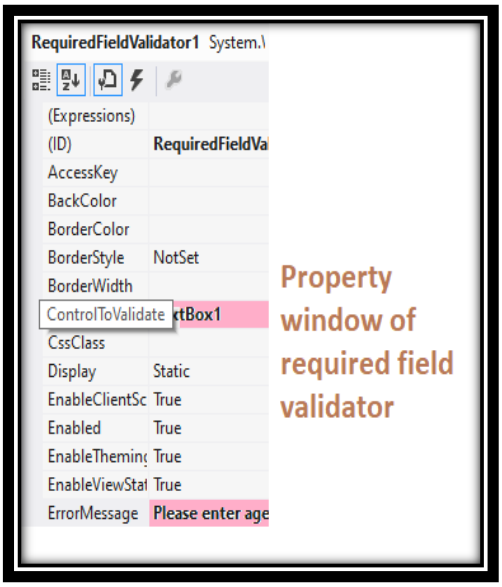




Coding:

Button1_CLick

Response.Write("Hello");



1 word Question Answer

Sr No.	Question	Answer
1.	……validation control is used when we want to check the value between specific range	Range Validator
02.	…….and ……property is used to set the value of range	Minumum Value Maximum Value

4) Regular Expression Validator:

- This Validator is used to check the value of a control against specified expression.
- Regular expression is a powerful tool which is used to check against specific type of values which may be string value, integer value, data value etc.
- Some examples of using regular expression Validator are:
  - to input mobile number in proper format
  - to input email address in valid format
- To make regular expression, following characters are specified below:

Sr No.	Character	Description
1.	\w	Any word character (letter, number or underscore)
2.	\d	Any digit
3.	\D	Any character that is not a digit
4.	\s	Any white space character
5.	\S	Any non-white space character
6.	[ ]	to specify range of characters or set of characters
7.	*	to specify zero or more no. of characters
8.	+	to specify one or more no. of charactes

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Some of the examples of regular expression are:

\d{10}:Mobile number

\d{5,}: Allows digit which can be minimum 5 with no maximum limit.

[A-D]: Allows any character between A to D

[A-D]*: Allows zero or any no. of characters between A to D

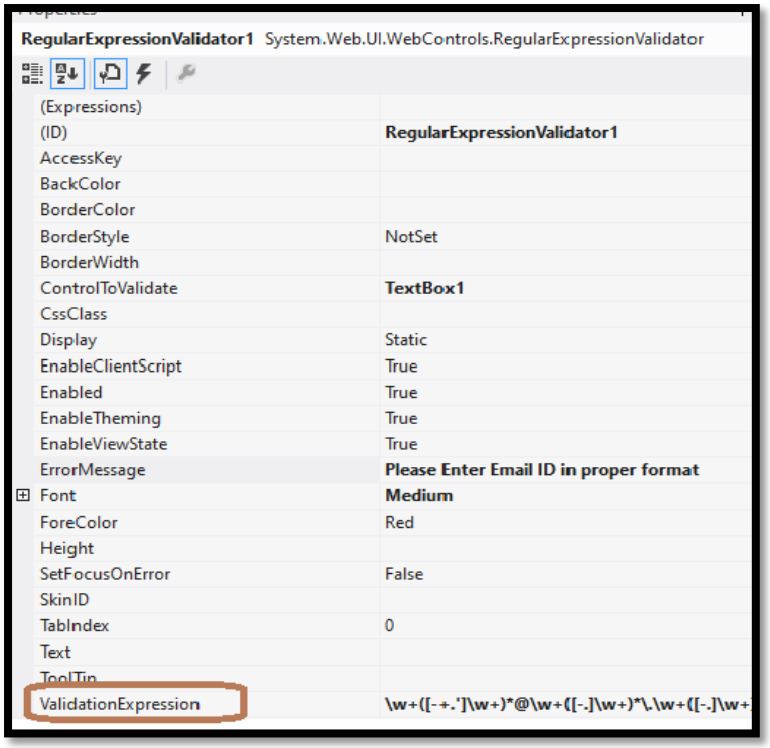
[A-D]+: Allows 1 or more characters between A to D.

Sr No.	Property	Description
1.	Error Message	This property is used to set the message which is displayed when Validator control fails.
2.	Control to validate	This property is used to set the control which you want as required field.
3.	Validation Expression	Regular expression for given field.

Example

Design View:

Property Window:



1 word Question Answer

Sr No.	Question	Answer
1.	……..control is used to check the value of control against specified expression	Regular Expression Validator
2.	.....property is used to add the expression?	validation expression

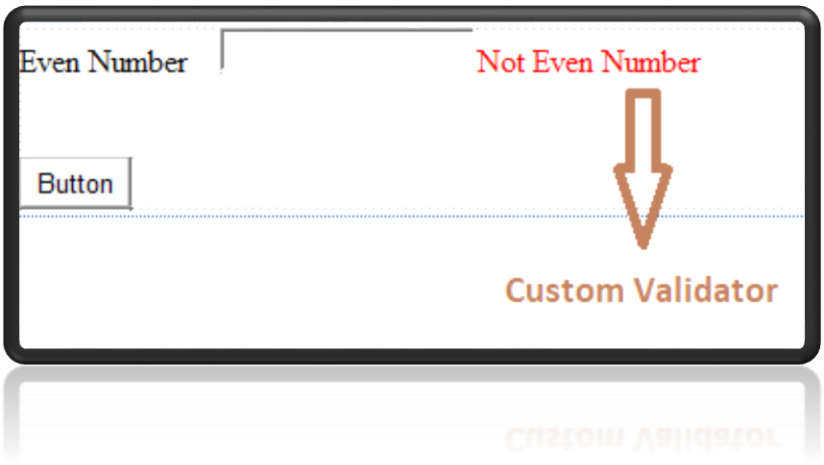
5) **Custom Validator:**

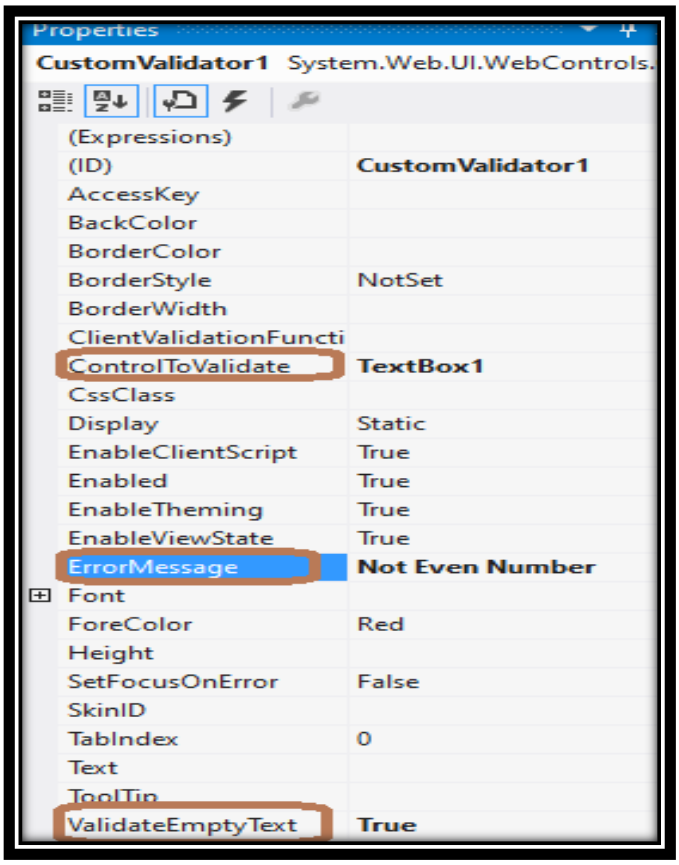
- Custom Validator control is used when the inbuilt validation controls cannot work.
- Using custom Validator, you can customize your checking with control.
- Custom Validator control allows you to specify your own functions at both sides.
    - Server side validation Function
    - Client side validation Function
- Server Side Validation Function:
    - This function is executed on server side. This is done when you click on button or some processing is done at server side.
    - The name of server side function is Server Validate function. This function has 2 arguments:
        - **object source:** This will give you the source name with which custom Validator control is connected.
        - **object args:** This will have 2 important properties:
            - args.Value: Will give you value of control
            - args.IsValid:can be set to true or false

Example

Sr No.	Property	Description
1.	Error Message	This property is used to set the message which is displayed when Validator control fails.
2.	Control to validate	This property is used to set the control which you want as required field.

Design View:




Properties
CustomValidator1 System.Web.UI.WebControls.
(Expressions)
(ID) CustomValidator1
AccessKey
BackColor
BorderColor
BorderStyle NotSet
BorderWidth
ClientValidationFuncti
ControlToValidate TextBox1
CssClass
Display Static
EnableClientScript True
Enabled True
EnableTheming True
EnableViewState True
ErrorMessage Not Even Number
Font
ForeColor Red
Height
SetFocusOnError False
SkinID
TabIndex 0
Text
ToolTip
ValidateEmptyText True

Coding:

CustomValidator1_ServerValidate

```
            if (args.Value==" ")
        {
            args.IsValid = false;
        }
        else
        {
       int num = Convert.ToInt32(args.Value);
       if (num % 2 == 0)
      args.IsValid = true;
       else
      args.IsValid = false;
    }
      }
```

Button1_Click

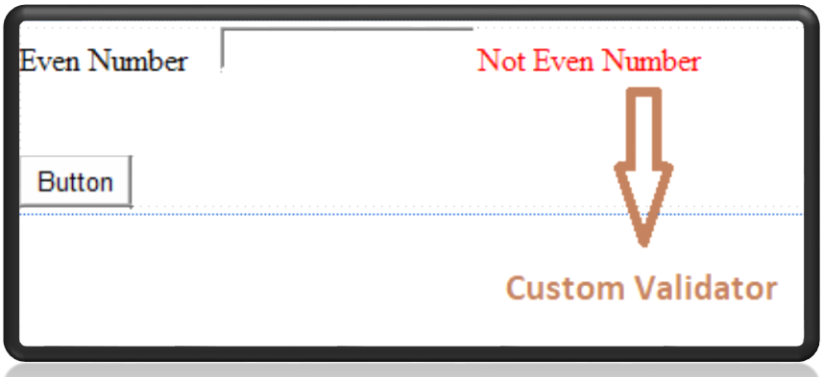
```
            if (Page.IsValid)
          Response.Write("Even Number");
        else
            Response.Write("Valdiaton Error");
```

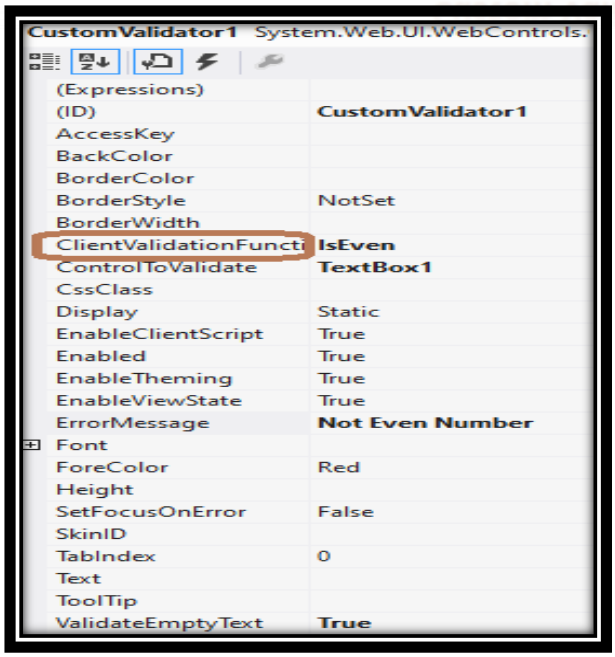
Client Side Validation Function:

- This function is executed on client side. Client side function can be written in JavaScript.
- The name of client side function should be specified in **Client Validation Function**.

Example

Design View:





Source:

```
<script language="javascript" type="text/javascript">
    function IsEven(source, args) {
        if (args.Value == "") {
            args.IsValid = false;
        }
        else {
            if (args.Value % 2 == 0)
                args.IsValid = true;
            else
                args.IsValid = false;
        }

        }
</script>
```

1 word Question Answer

Sr No.	Question	Answer
1.	Custom Validator is used when inbuilt controls does not work (T/F)	True
2.	Which event is fired or what is the name of event to write the code on server side custom validation.	ServerValidate
3.	args.IsValid means?	It is Boolean property having two value (True/False) depending upon the condition

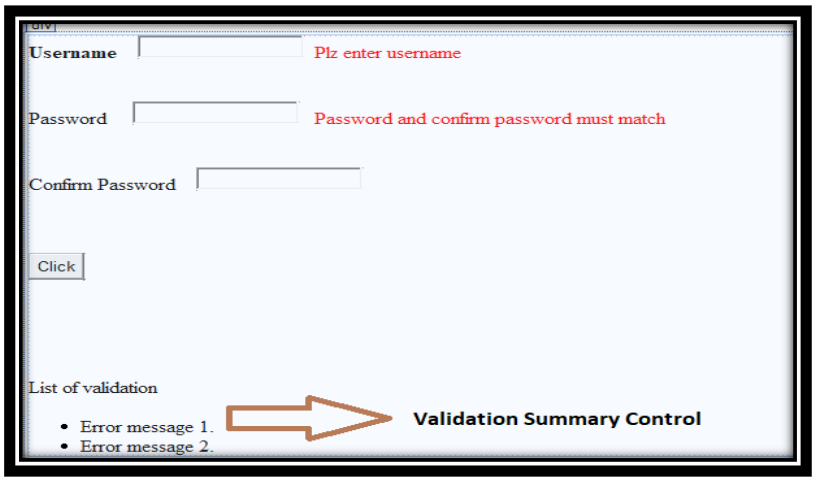
6) Validation Summary Control:

- This control is used to show summary of all the Validator controls which are included in particular web form.
- It is very useful when you have large web form which has more number of controls, may not be visible in single screen

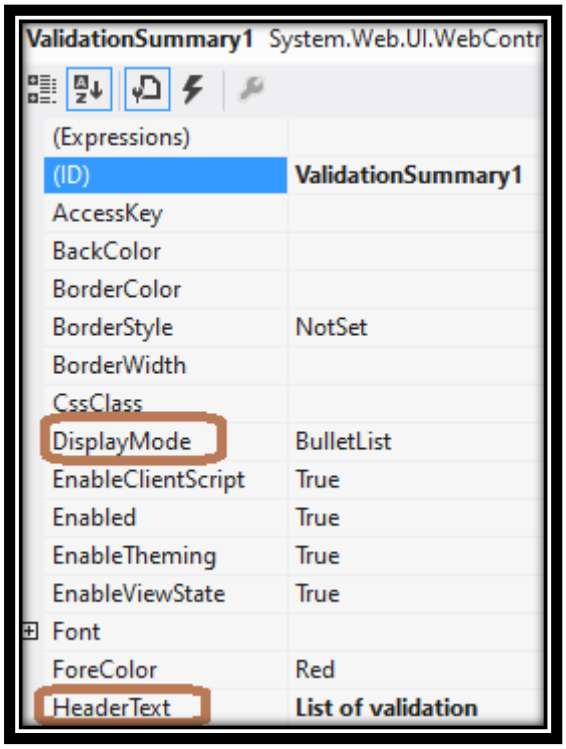
Sr No.	Property	Description
1.	Header Text	This property is used to give heading to the validation summary control
2.	Display Mode	• Bullet List • List • Paragraph

Example:

Design View:



Property Window:

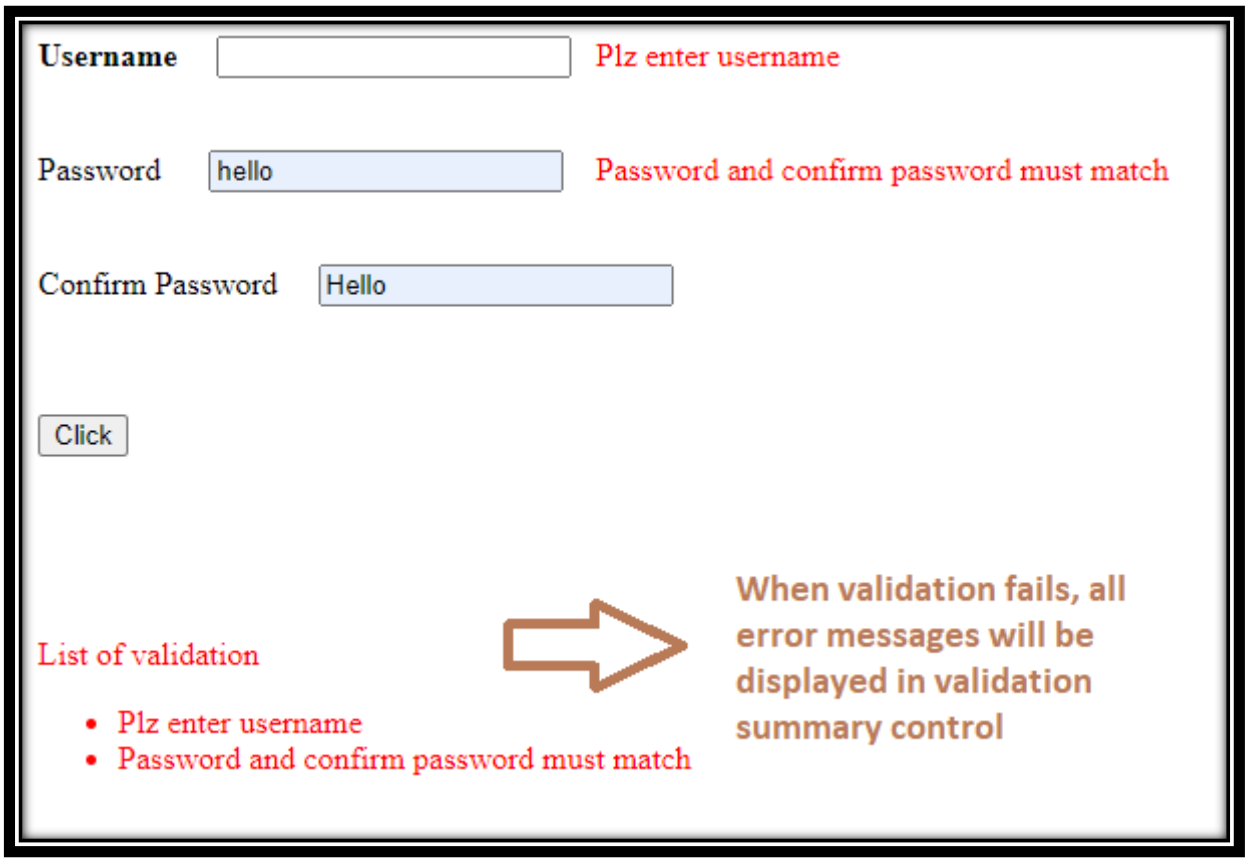


Coding:

<u>Button1_Click</u>

```
if(Page.IsValid==true)
        Response.Write("Hello");
```

Output:



1 word Question Answer

Sr No.	Question	Answer
1.	……control is used to show summary of all the Validator controls which are included in particular web form	Validation Summary Control

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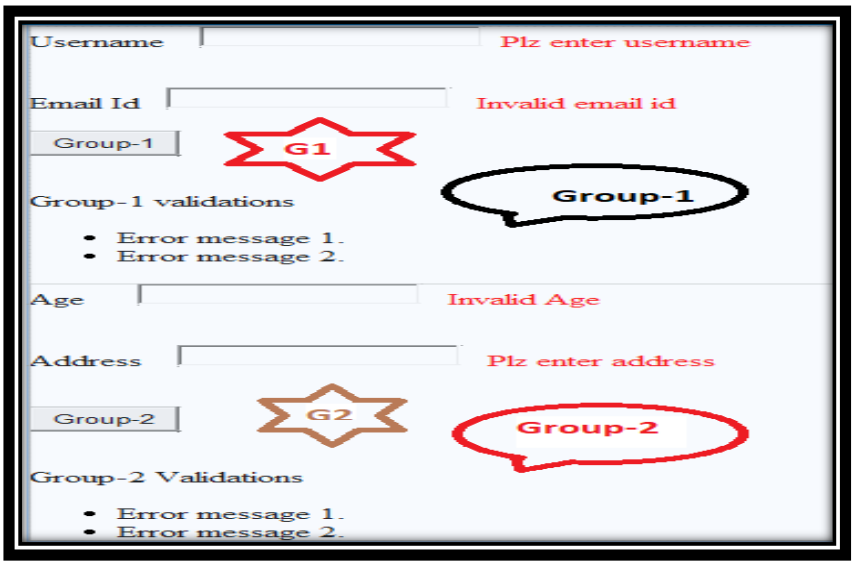
7) Validation Group:

- All validation controls has a property called "IsValid". If all validation controls "IsValid" property is true then only "IsValid" property of "Page" class is set to true, which allows you to execute your server side event code.
- So, in this case, even if one of the validation control is not valid, the whole page will be treated as "Invalid=false", which won't allow to process your event code written in code behind file.
- But in case, if you have two or more different group of validation controls, if one group of validation control is valid, related button should be allowed to click and its event should be executed. At that time may be one of other validation group control may be incorrect.
- You need to set **"ValidateGroup" Property**

Example:

Design View



Step 2: Please set validation controls property of Group 1 to G1

"ValidateGroup" – G1

Step 3: Please set validation summary control property of group 1 to G1

"ValidateGroup" – G1

Step 4: Please set button1 property of group-1 to G1

"ValidateGroup" – G1

Step 5: Please set validation controls property of Group 2 to G2

"ValidateGroup" – G2

Step 6: Please set validation summary control property of group 1 to G2

"ValidateGroup" – G2

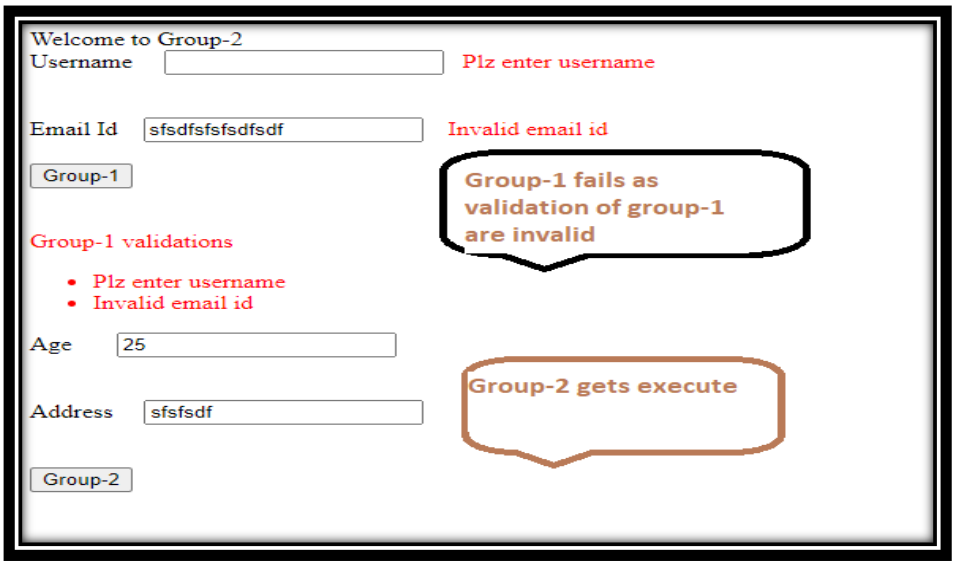
Step 7: Please set button2 property of group-2 to G2

"ValidateGroup" – G2

```
protected void Button1_Click(object sende
{
    Response.Write("Welcome to group-1");
}

protected void Button2_Click(object sende
{
    Response.Write("Welcome to Group-2");
}
```

Output:



1 word Question Answer

Sr No.	Question	Answer
1.	Which property must be set to perform validation on particular group of controls	valdiateGroup